

Draft - expert review pending

This unit is complete and passes the platform's automated checks, but no reviewer has read it yet. Treat its sources and claims with care.

Unit 1 assessment

Unit summary

Environmental science studies the natural environment and the human-environment interaction across five scope areas: air, water, land, living things and human systems. The natural environment is organised into four overlapping spheres, atmosphere, hydrosphere, lithosphere and biosphere, and people live inside the biosphere, not beside it. Environmental problems are anthropogenic: human activities change the environment through the four-step nexus of activity, environmental response, harm returning and human response, as Sindh's waterlogging and salinity shows. Issues appear at global, regional and local scales, and the scales connect in both directions. Ecosystems provide services people depend on, face five main threats, and are conserved by protected areas, sustainable use, pollution control and restoration. Natural resources sort into renewable and non-renewable, and the management question is rate for the first and replacement for the second. Individual responsibility is the five-step path: notice, learn, reduce, reuse and recycle, influence others.

Summative assessment

Multiple-choice questions (MCQs)

1. In this unit's account, environmental science is best defined as the study of:
(Remember) a) wild animals and their protection b) the natural environment and human interactions with it c) weather and climate patterns only d) recycling and waste management
2. A teacher in Sukkur boils water for tea using natural gas. The gas, before extraction, belonged to which sphere? (Apply) a) the biosphere b) the

lithosphere c) the atmosphere d) the hydrosphere

3. "Anthropogenic" is best defined as: *(Remember)* a) caused by people b) occurring naturally c) related to animals d) found only in cities
4. Waterlogging and salinity in lower Sindh most directly resulted from: *(Understand)* a) over-fishing in the lakes b) canal seepage and over-watering raising the water table c) a rise in sea level alone d) the planting of mangroves
5. The winter smog that builds at the same time over several South Asian countries is best classified as: *(Apply)* a) a local issue b) a regional issue c) a global issue d) a personal issue
6. The two kinds of parts in every ecosystem are: *(Remember)* a) plants and animals b) living and non-living c) renewable and non-renewable d) producers and consumers
7. Which of these is an ecosystem service of the Indus Delta mangroves? *(Understand)* a) they increase the salinity of farmland b) they soften storm waves and protect coastal villages c) they consume the fish stocks of the creeks d) they block the flow of the river
8. A farmer pumps groundwater faster than recharge can replace it. In this situation, the groundwater is behaving as: *(Apply)* a) a renewable resource in normal use b) a non-renewable resource in practice c) an ecosystem service d) an abiotic factor that cannot run out
9. In the individual action path, the step that is strongest because "waste never made never needs managing" is: *(Understand)* a) reuse b) recycle c) reduce d) notice
10. Buying one large tin of biscuits instead of twelve small wrapped packets is an example of which step? *(Apply)* a) reduce b) recycle c) influence others d) notice

Restricted-response questions (RRQs)

1. In two sentences, explain why a city street counts as an environment.
(Understand)

2. Trace a cup of tea through the four spheres in four short phrases, one per sphere. (*Apply*)
3. Name the four steps of the nexus, and state in one sentence which step the white salt crust on a Sindh field belongs to. (*Understand*)
4. Give two reasons why "smog in Lahore is only a local issue" is the wrong classification. (*Analyze*)
5. Explain why a global problem such as climate change shows up locally. Use one Pakistani example in no more than three sentences. (*Analyze*)
6. State the difference between renewable and non-renewable resources, with one example of each from Pakistan. (*Understand*)
7. Why does groundwater in lower Sindh behave like a non-renewable resource? Name the rate that decides the answer. (*Analyze*)
8. Match habitat loss, overuse and pollution each to one conservation tool, and justify one of your three matches in a sentence. (*Analyze*)
9. In two sentences, explain why reduce comes before reuse and recycle in the action path. (*Understand*)
10. A colleague says, "Individual action is pointless next to factory pollution." Give the honest two-part answer this unit gives. (*Analyze*)

Extended-response questions (ERQs)

1. Using the four-sphere picture, explain what environmental science studies and why it matters for a teacher in Sindh. Support your answer with at least two real examples. (*Analyze*)
2. Choose one anthropogenic environmental problem you have personally seen in Pakistan. Trace it through the four nexus steps, and finish by naming who bears most of its harm and why. (*Analyze*)
3. Your practicum school sits inside a real ecosystem and uses real resources. Identify the ecosystem, name two services it provides, and analyse one resource the school uses: its class, its rate of use, and one sustainable rule for it. (*Evaluate*)
4. From a real one-week use-and-waste log, analyse your single largest avoidable use or waste, and design the one change you would keep, naming one condition under which it would fail. (*Create*)
5. Take one local environmental issue, such as a burning waste heap or a drying lake, and analyse its connections to the regional and global scales.

Conclude by proposing who should act at each scale, and what one individual teacher can actually do. (*Evaluate*)

Answers and marking guidance

MCQ answer key

1. b - environmental science studies the environment and the human interaction with it; the other options each describe only one small part.
2. b - natural gas is extracted from rock layers, which is the lithosphere.
3. a - anthropogenic means caused by people, from the Greek for human.
4. b - canal seepage and over-watering raised the water table and drew salts to the surface; that is the worked example in Topic 1.2.
5. b - smog shared across several countries at once is regional; its sources and effects cross borders.
6. b - the living (biotic) and non-living (abiotic) parts; producers and consumers are only two roles within the living part.
7. b - the mangroves' roots soften storm waves and hold the delta soil, protecting coastal villages.
8. b - when use exceeds recharge, a renewable resource behaves as non-renewable; the rate of pumping decides.
9. c - reduce; waste never made needs no collection, separation or processing.
10. a - reduce; the tin replaces many wrappers, so the waste is never created.

RRQ model answers and mark schemes

1. Model: A street is part of the biosphere people live in; its air, dust, drainage and shade are the surroundings that affect living things, so it is an environment. **Marks (2):** 1 for placing people inside the biosphere; 1 for naming a street-level environmental element (air, dust, drainage).
2. Model: hydrosphere, the boiled water; lithosphere, the gas that burned; biosphere, the tea leaf that grew; atmosphere, the steam and the rain that watered the leaf. **Marks (4):** 1 per correct sphere with its item.
3. Model: activity, environmental response, harm returning, human response; the salt crust is the environmental response (or the returning harm if framed

as lost yield; accept either with a reason). **Marks (3):** 1 for the four steps in order; 1 for placing the crust; 1 for the one-sentence reason.

4. Model: first, the sources are shared, vehicle and industrial emissions and crop burning occur across the region; second, the air itself crosses borders, so one city's emissions add to another's. **Marks (2):** 1 per reason.
5. Model: global greenhouse gases raise temperatures everywhere; locally this appears as heatwaves, such as Karachi's, or changed rainfall. **Marks (2):** 1 for the mechanism; 1 for a specific Pakistani example.
6. Model: renewable resources renew on a human timescale if not overused, such as Indus water or lake fish; non-renewable resources are fixed stocks, such as Thar coal or natural gas. **Marks (2):** 1 for the distinction; 1 for both examples.
7. Model: recharge is slow because rainfall is thin, and pumping exceeds it, so the water table falls; the rate of pumping versus the rate of recharge decides. **Marks (2):** 1 for the pumping-versus-recharge explanation; 1 for naming the deciding rate.
8. Model: habitat loss to protected areas; overuse to sustainable use rules such as catch limits; pollution to pollution control. Justification example: a catch limit slows overuse because it caps the take below renewal. **Marks (3):** 1 per correct match; the justification must link tool to threat.
9. Model: reduce prevents the waste from existing, while reuse and recycle only manage waste that has already been made; preventing beats managing. **Marks (2):** 1 for the prevention-versus-management contrast; 1 for clarity.
10. Model: part one, individual action is not a substitute for policy, and factories and systems matter more than habits; part two, individual action is not worthless, because systems are fed by households and a teacher multiplies habits through a class. **Marks (2):** 1 per half, both needed.

ERQ rubrics

1. **Analytic rubric.** *Sphere reasoning (0-3, Analyze: traces at least two examples through the spheres to show people are inside the environment) · Definition and scope used (0-2) · Pakistani evidence (0-3, Apply: specific, real examples such as city air, the Indus, Thar) · Clarity and register (0-2).*
Total 10.

2. **Analytic rubric.** *Nexus traced in order (0-4, Analyze: all four steps with the specific activity and returning harm) · Harm-bearer named with reason (0-2, Evaluate) · Specificity of the chosen problem (0-2) · Clarity (0-2). Total 10.*
3. **Analytic rubric.** *Ecosystem identified correctly (0-2) · Two services with concrete links to the school (0-2) · Resource analysed (0-4, Analyze: class, rate and rule all addressed) · Clarity (0-2). Total 10.*
4. **Analytic rubric.** *Log item is real and specific (0-2) · Five steps applied honestly (0-3) · Change designed with a realistic failure condition (0-3, Create: the rule is workable and the failure condition names a rate or a pressure) · Clarity (0-2). Total 10.*
5. **Analytic rubric.** *Local issue described specifically (0-2) · Regional connection analysed (0-2, Analyze) · Global connection analysed (0-2, Analyze) · Actors proposed per scale with the teacher's part realistic (0-3, Evaluate) · Clarity (0-1). Total 10.*

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